

46 YEARS OF EXCELLENCE | SRAMAYEVA JAYATHE | NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 20/01/2026**
Time: 3 Hrs **GTM - 20** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. The magnetic field in a plane EM wave which is moving in the free space is given by-

$B = (100\ \mu T)\sin[(2\times10^{15}\ s^{-1})(t - x/c)]\ \hat{j}$

The equation for electric field is-

A) $E = 100\ \mu N/C\ \sin[(2\times10^{15}s^{-1})\ (t-x/c)]\ (-\hat{k})$

B) $E=3\times10^{10}\mu N/C\ \sin[(2\times10^{15}s^{-1})\ (t- x/c)]\ (-\hat{k})$

C) $E=3\times10^{10}\mu N/C\ \sin[(2\times10^{15}\ s^{-1})\ (t - x/c)]\ \hat{k}$

D) $E = 100\ \mu N/C\ \sin[(2\times10^{15}\ s^{-1})\ (t - x/c)]\ \hat{k}$

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02. Van der Waal's gas equation is $\left(P+\frac{a}{V^2}\right)(V-b)=RT$ where symbols have their usual meaning. The dimensions of quantity a/b are –

A) $M L^4 T^{-2}$ B) $ML^5 T^{-2}$ C) $M L^3 T^{-2}$ D) $M L^2 T^{-2}$

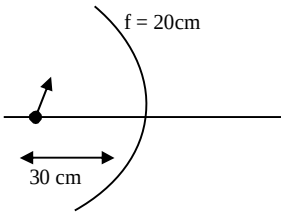
03. In the given diagram mass of pulley is m, radius is R and radius of gyration is k. An inextensible string is wrapped tightly on it and a ball of mass 'm' is attached to the free end. After falling a height h the string becomes tight. The kinetic energy of the ball just after string becomes tight is



A) mgh B) $\frac{mgh}{\left(1+\frac{R^2}{k^2}\right)}$

C) $\frac{mgh}{\left(1+\frac{k^2}{R^2}\right)^2}$ D) none of these

04. A point object is projected from a point on the principal axis of a concave mirror of focal length 20 cm with a speed of $\sqrt{5}$ m/s at an angle of $\tan^{-1}(2)$ with the principal axis as shown in figure. Choose the correct option. (consider the moment just after projection)



A) magnitude of longitudinal magnification is 1

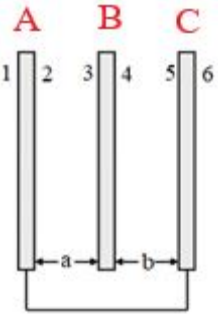
B) image speed is $5\sqrt{2}m/s$

C) image velocity makes an angle of 30^0 with the principal axis

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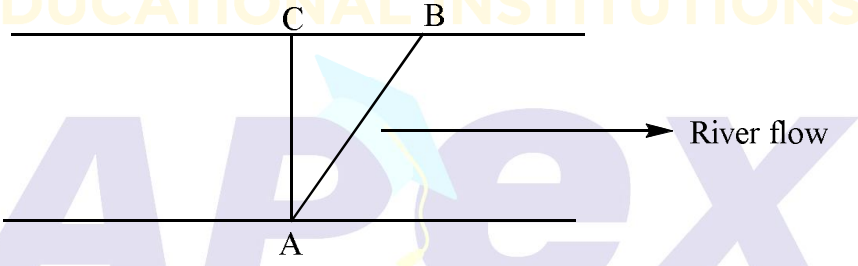
D) relative speed of image with respect to object is $\sqrt{61}\,m/s$

05. Three identical metallic plates are kept parallel to one another at a separation of a and b. The outer plates are connected by a thin conducting wire and a charge Q is placed on the central plate. The force on a point charge q placed between the plates A and B is



A) $\frac{aQq}{A(a+b)\in_0}$ B) $\frac{bQq}{A(a+b)\in_0}$ C) $\frac{bQq}{A(a-b)\in_0}$ D) $\frac{aQq}{A(a-b)\in_0}$

06. Width of a river is 60 m. A swimmer wants to cross the river such that he reaches from A to B directly. Point B is 45 m ahead of line AC (perpendicular to river) Assume the speed of river and speed of swimmer in river are equal. Swimmer must try to swim at angle θ with line AC. If the value of θ is



A) 16^0 B) 37^0 C) 53^0 D) 0^0

07. Two point charges + 8q and – 2q are located at x = 0 and x = L respectively. The location of a point on the x-axis at which the net electric field due to these two point charges is zero is

A) 8L B) 4 L C) 2 L D) L/4

08. How much electrical energy in kilo-watt hour is consumed in operating ten bulbs each rated as 50 watt for 10 hours per day in a month of 30 days?

A) 1500 B) 15000 C) 15 D) 150

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09. The displacement equation of a particle is x = 3 sin 2t + 4cos 2t. The amplitude & maximum velocity will be respectively (All are taken in S.I. units)

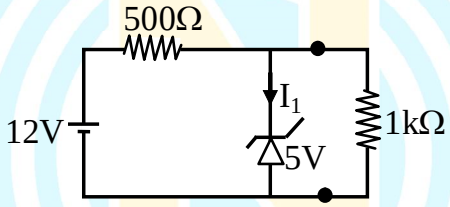
A) 5, 10 B) 3, 2 C) 4, 2 D) 3, 4

10. The relation between magnetic susceptibility χ_m and relative permeability μ_r, is__

A) $\chi_m = \mu_r$ B) $\chi_m - 1 = \mu_r$

C) $\mu_r = 1 + \chi_m$ D) $\mu_r = 1 - \chi_m$

11. The current flowing through the zener diode in figure is –



A) 2 mA B) 7 mA C) 9 mA D) 5 mA

12. In an adiabatic process, which of the following statements are false

I) The molar heat capacity is infinite

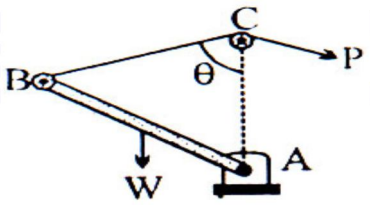
II) Work done by the gas equals the increase in internal energy

III) The molar heat capacity is zero

IV) The internal energy of the gas decreases as the temperature increases

A) I,II B) I,II,III C) I,II,IV D) I,II,III,IV

13. A uniform rod AB of weight W is movable in a vertical plane about a smooth hinge at A, and is sustained in equilibrium by a force P acting along a string BCP passing over a smooth peg C as shown. AC being vertical. If AC be equal to AB, then the force P is

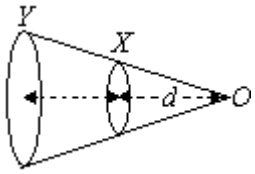


A) $W/\cos\theta$ B) $W\cos\theta$ C) $W/\sin\theta$ D) $W\sin\theta$

14. Two circular coils X and Y have equal number of turns and carry equal currents in the same sense and subtend same solid angle at point O. If the smaller coil X

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is midway between O and Y, then if we represent the magnetic induction due to bigger coil Y at O as B_Y and that due to smaller coil X at O as B_X, then



A) $\frac{B_Y}{B_X}=1$ B) $\frac{B_Y}{B_X}=2$ C) $\frac{B_Y}{B_X}=\frac{1}{2}$ D) $\frac{B_Y}{B_X}=\frac{1}{4}$

15. Which of the following statements are true regarding Bohr's model of hydrogen atom –

(I) Orbiting speed of electron decreases as it shifts to discrete orbits away from the nucleus

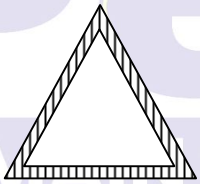
(II) Radii of allowed orbits of electron are proportional to the principal quantum number

(III) Frequency with which electrons orbits around the nucleus in discrete orbits is inversely proportional to the cube of principal quantum number

(IV) Binding force with which the electron is bound to the nucleus increases as it shifts to outer orbits. Select correct answer using the codes given below

A) I and III B) II and IV C) I, II and III D) II, III and IV

16. A rod of mass 3 kg and of length 3m is bent in the form of an equilateral triangle. The moment of inertia of the triangle about a vertical axis perpendicular to plane and passing through centre of mass is-



A) 1/6 kg-m^2 B) 1/3 kg-m2 C) 5/12 kg-m^2 D) 1/2 kg-m^2

17. A small object of height 0.5 cm is placed 30cm in front of a convex surface of glass (μ=1.5) of radius of curvature 10 cm. Find the height of the image formed in glass.

A) 2 cm B) 1 cm C) 3 cm D) 4 cm

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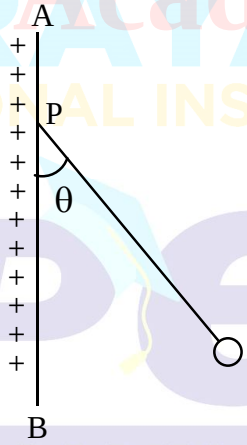
18. Match the dimensions given in the column II with the corresponding physical quantities in the column I.

Column I	Column II
(a) Resistance	(p) $\left[ML^2 T^{-3} I^{-1}\right]$
(b) Potential difference	(q) $\left[ML^{-1} T^{-1}\right]$
(c) Viscosity	(r) $\left[M^{-1} L^3 T^{-2}\right]$
(d) Universal Gravitational constant	(s) $\left[ML^2 T^{-3} I^{-2}\right]$

A) a-r,b-s,c-q,d-p B) a-q,b-p,c-s,d-r

C) a-r,b-s,c-p,d-q D) a-s,b-p,c-q,d-r

19. An infinite plane with uniformly distributed positive charge has surface charge density σ C/m^2 is placed in air. A small metallic sphere S of mass m and charge +Q is attached to a thread and tied to a point P on the sheet AB. The angle which PS makes with the plane AB is given by ($\in_0$ = permittivity in vacuum, $\in_r$ = relative permittivity of air)



A) $\tan^{-1}\frac{Q\sigma}{\in_0 \in_r mg}$ B) $\cot^{-1}\frac{Q\sigma}{2\in_0 \in_r mg}$

C) $\tan^{-1}\frac{Q\sigma}{2\in_0 \in_r mg}$ D) $\cot^{-1}\frac{Q\sigma}{\in_0 \in_r mg}$

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20. The radiation emitted, when an electron jumps from n = 3 to n = 2 orbit is a hydrogen atom, falls on a metal to produce photoelectron. The electrons from the metal surface with maximum kinetic energy are made to move perpendicular to a magnetic field of $\frac{1}{320}$ T in a radius of 10^{-3}m. Find the work function of metal-

A) 1.03 eV B) 1.89 eV C) 0.86 eV D) 2.03 Ev

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. In the figure shown, the heavy cylinder (radius R) resting on smooth surface separates two liquids of densities 2 ρ and 3ρ. The height h for the equilibrium of cylinder must be $R\sqrt{\frac{n}{2}}$. The value of n is___.



22. A uniform rope of mass m and length L is hung freely from stationary ceiling. If the cross sectional area of rope is A and Young`s modulus Y, then net elongation in the rope due to its own weight is $\frac{mgL}{\beta AY}$, then the value of β is

23. A slit of width 'a' is illuminated by white light. The first minimum for red light (λ = 6500Å) will fall at θ = 30°, the value of 10a is (in μm)-

24. At NTP the density of a gas is 1.3 kg/m^3 and the velocity of sound propagation in the gas is 330 m/s. The degree of freedom of gas molecule is (atmospheric pressure is 1×10^5 Pa)

25. A particle moving in a straight line covers half the distance with speed of 3 m/s. The other half of the distance is covered in two equal time intervals with a speeds of 4.5 m/s and 7.5 m/s, respectively. The average speed of the particle during this motion is (in m/s)

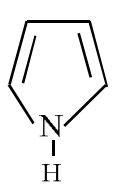
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SECTION – I
(SINGLE CORRECT ANSWER TYPE)

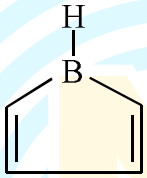
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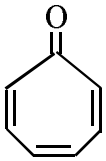
26. Designate whether each of the following compounds is aromatic or anti aromatic



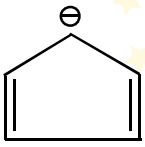
(a)

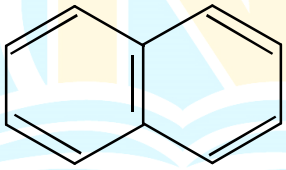


(b)

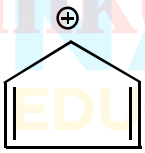


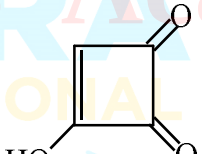
(c)


(d)


(e)


(f)


(g)



(h)

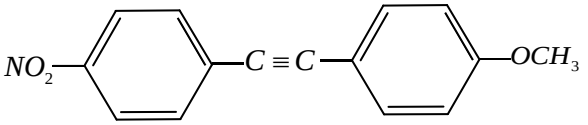
A) e, g aromatic and a, b, c, d, f, h anti aromatic

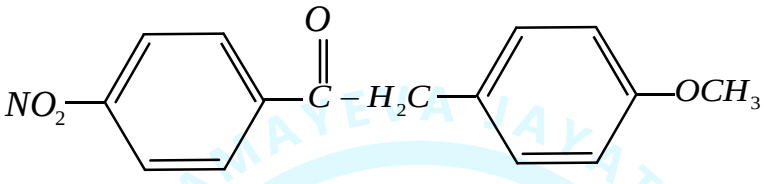
B) b, e, f, g aromatic and a, c, d, h anti aromatic

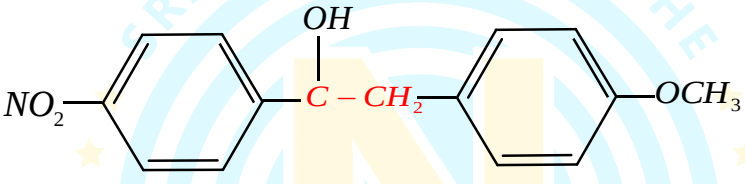
C) a, b, c, d aromatic and e, f, g, h anti aromatic

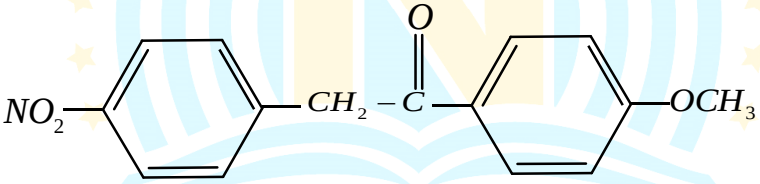
D) a, c, d, e, h aromatic and b, f, g anti aromatic

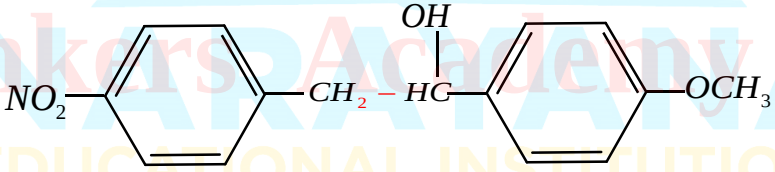
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27.  $\xrightarrow{HgSO_4/H_2SO_4}$ A(Major product); A is

A) 

B) 

C) 

D) 

28. The property/properties that show irregularity in first four elements of group-17 is/are

1) Electronegativity 2) Bond dissociation enthalpy

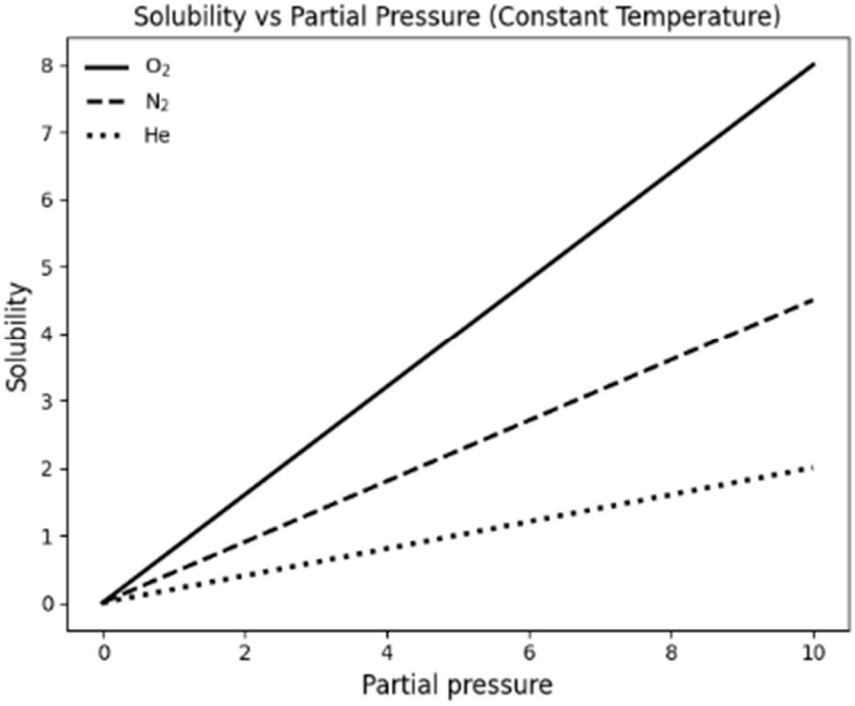
3) Melting point 4) Boiling point

Choose the correct answer from the options given below:

A) 2 and 4 only B) 1 and 3 only C) 2 only D) 1,2,3 and 4

29. Molar solubility of helium, nitrogen and oxygen are plotted against partial pressure of the gas at constant temperature

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Assume Henry's law is obeyed throughout the pressure range and gases behave ideally.

Which **one** of the following statements is **correct**?

A) The Henry's law constants satisfy $k_H(O_2) > k_H(N_2) > k_H(He)$

B) At the same partial pressure, the number of moles of He dissolved per litre is maximum because it has the smallest molar mass.

C) If the temperature is increased slightly, the slopes of all three lines will increase.

D) If partial pressure (P) is plotted on the y-axis and molar solubility (S) on the x-axis, the slope of each straight line will be equal to the corresponding Henry's law constant.

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30. Consider the following statements related to atom and its structure:

I. In Bohr's model of a hydrogen-like atom, the speed of an electron in a circular orbit decreases with increase in the principal quantum number n.

II. The term orbital speed of an electron has no physical meaning in quantum mechanics and is therefore avoided while describing atomic orbitals.

III. For hydrogen-like species, the ratio of frequencies of photons emitted in the transitions $3 \rightarrow 2$ and $2 \rightarrow 1$ is independent of the atomic number Z.

IV. The explanation of blackbody radiation and the photoelectric effect both require the assumption that energy exchange between matter and radiation occurs in discrete packets.

Which of the following options is **correct**?

A) Only I and II are correct

B) Only I, II and III are correct

C) Only II, III and IV are correct

D) All four statements are correct

31. In an adiabatic process, no transfer of heat takes place between system and surroundings. Choose the correct option for free expansion of an ideal gas under adiabatic condition from the following.

A) $q = 0, \Delta T \neq 0, w = 0$ B) $q \neq 0, \Delta T = 0, w = 0$

C) $q = 0, \Delta T = 0, w = 0$ D) $q = 0, \Delta T < 0, w \neq 0$

32. Benzene and toluene form nearly ideal solutions. At 20° C, the vapour pressure of benzene is 75 torr and that of toluene is 22 torr. The partial vapour pressure of benzene at 20° C for a solution containing 78 g of benzene and 46 g of toluene in torr is

A) 50 B) 25 C) 37.5 D) 53.5

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33. Blister copper is produced by reaction of copper oxide with copper sulphide.

$2Cu_2O + Cu_2S \rightarrow 6Cu + SO_2$, When $2.86\times10^3\,g$ of Cu_2O and $4.77\times10^3\,g$ of Cu_2S are used for reaction, the mass of copper produced is _____g.

(nearest integer)

(atomic mass of Cu = 63.5 a.m.u , S = 32.0 a.m.u , O = 16 a.m.u)

A) 3810 B) 1905 C) 1270 D) 3900

34. The precipitate of $CaF_2\left(K_{sp}=1.7\times10^{-10}\right)$ is obtained when equal volumes of the following are mixed

A) $10^{-4}\,M\,Ca^{2+} + 10^{-4}\,M\,F^-$ B) $10^{-2}\,M\,Ca^{2+} + 10^{-3}\,M\,F^-$

C) $10^{-5}\,M\,Ca^{2+} + 10^{-3}\,M\,F^-$ D) $10^{-3}\,M\,Ca^{2+} + 10^{-5}\,M\,F^-$

35. Among $BF_3, CHCl_3, CO_2, PCl_5, XeF_4, XeF_2, CCl_4$. the hybridization of molecule with non-zero dipole moment is

A) sp^2 B) sp C) d^2sp^3 D) sp^3

36. **Statement-I:** Cinnamaldehyde reacts with both Tollen's reagent and Fehling's reagent

Statement-II: Both aliphatic and aromatic aldehydes give positive test with Fehling's Reagent

A) **Statement-I** is incorrect but **Statement-II** is correct

B) **Statement-I** is correct but **Statement-II** is incorrect

C) **Statement-I** and **Statement-II** are incorrect

D) Both **Statement-I** and **Statement-II** are correct

37. Identify the correct statement among the following

A) All naturally occurring amino acids are optically inactive

B) Isoleucine has one chiral center

C) Aspartic acid is the only amino acid among 20 alpha amino acids that contains a-COOH group in the side chain

D) Cysteine and methionine are Sulphur containing amino acids

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38. The correct order of basic nature in aqueous solution for the following bases $NH_3, CH_3-NH_2, (CH_3)_2NH, (CH_3)_3N$, is

A) $NH_3 < (CH_3)_3N < CH_3-NH_2 < (CH_3)_2NH$

B) $NH_3 < CH_3-NH_2 < (CH_3)_2NH < (CH_3)_3N$

C) $CH_3-NH_2 < NH_3 < (CH_3)_2NH < (CH_3)_3N$

D) $(CH_3)_3N < (CH_3)_2NH < CH_3-NH_2 < NH_3$

39. Given two statements below:

Statement I: In Cl_2, molecule the covalent radius is double of the atomic radius of chlorine.

Statement II: Radius of anionic species is always greater than their parent atomic radius.

Choose the most appropriate answer from options given below:

A) Both statement I and statement II are correct.

B) Both statement I and statement II are incorrect.

C) Statement I is correct but statement II is incorrect.

D) Statement I is incorrect but statement II is correct.

40. Assertion (A): The complex $\left[Fe(CN)_6\right]^{3-}$ exhibits paramagnetism, despite the fact that electrons occupy only the lower energy t_{2g} orbitals.

Reason (R): Occupation of only the t_{2g} orbitals in an octahedral complex necessarily implies complete pairing of electrons.

A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

B) Both (A) and (R) are true, and (R) is NOT the correct explanation of (A).

C) (A) is true, but (R) is false

D) (A) is true, but (R) is false

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41. List-1 (Color) | List-2 (Element)

A) Prussian blue — 1) $\left[Fe(CN)_5 NOS\right]^{4-}$

B) Violet — 2) $Fe_4\left[Fe(CN)_6\right]_3$

C) Blood red — 3) $\left[Fe(SCN)\right]^{2+}$

D) White — 4) AgCl

The correct match is

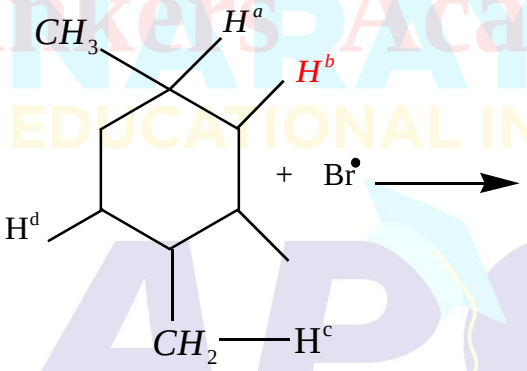
A) A-4,B-1,C-3,D-2 B) A-3,B-1,C-2,D-4

C) A-1,B-2,C-3,D-4 D) A-2,B-1,C-3,D-4

42. 0.246g of an organic compound on complete combustion gave 0.198g of carbon dioxide and 0.1014g of water, then the percentage composition of carbon and hydrogen in the compound respectively

A) 4.58, 21.95 B) 21.95, 4.58 C) 45.8, 2.195 D) 2.195, 45.8

43.

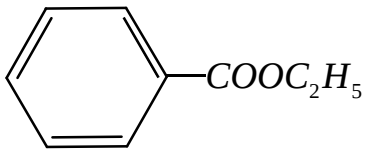


$Br^{\bullet}$ will abstract which of the designated hydrogen most readily.

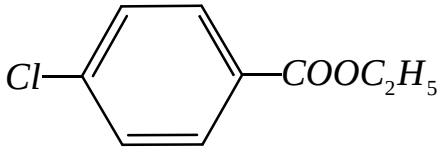
A) a B) b C) c D) d

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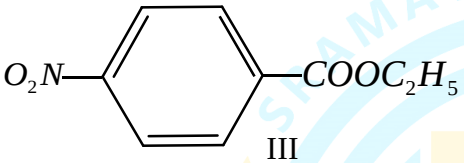
44. The decreasing order of ease of alkaline hydrolysis for the following esters is



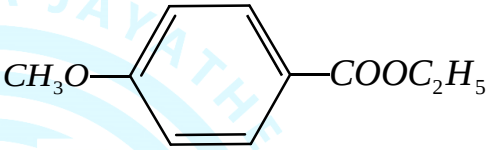
I



II



III



IV

A) III>II>IV>I B) III>II>I>IV C) IV>II>III>I D) IV>I>II>III

45. The shape/structure of $\left[XeF_5\right]^-$ and XeO_3F_2, respectively are

A) pentagonal planar and trigonal bipyramidal

B) octahedral and square pyramidal

C) trigonal bipyramidal and pentagonal planar

D) trigonal bipyramidal and trigonal bipyramidal

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. The spin-only magnetic moment value of M^{3+} ion (in gaseous state) from the pairs Cr^{3+}/Cr^{2+}, Mn^{3+}/Mn^{2+}, Fe^{3+}/Fe^{2+} and Co^{3+}/Co^{2+} that has negative standard electrode potential is_____ (in B.M.) [Nearest integer]

47. At what pH, given half cell $MnO_4^-\left(0.1M\right), H^+_{(aq)} \mid Mn^{2+}\left(0.001M\right)$ will have electrode potential of 1.282 V?____

Given $E^0_{MnO_4^-, H^+ \mid Mn^{2+}} = 1.54V, \dfrac{2.303RT}{F} = 0.059V$

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48.

$$C_2H_2 \xrightarrow[tube]{hot\ Fe} (A) \xrightarrow[AlCl_3,\Delta]{CH_3CH_2CH_2Cl} (B) \xrightarrow{O_2} (C) \xrightarrow{H^+/H_2O} (D)$$

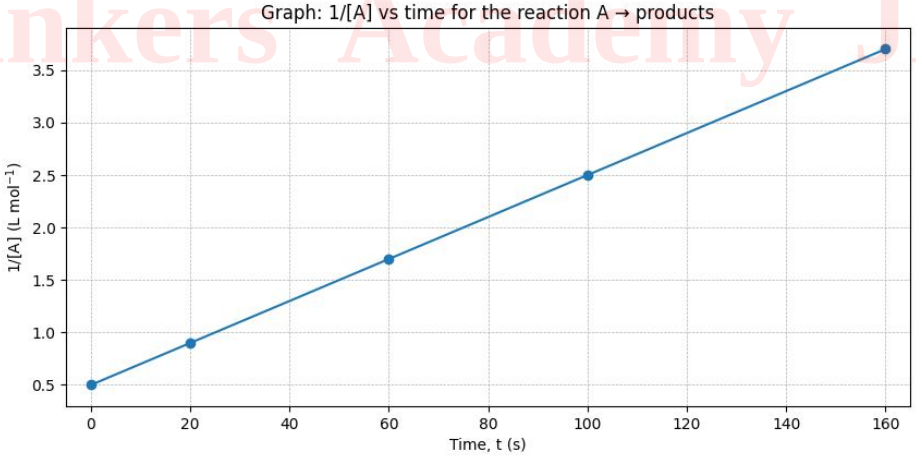
$$(E) \xleftarrow{Conc.HNO_3}$$

The gram molecular weight of compound (E) is_____

49. 4.0 moles of argon and 5.0 moles of PCl_5 are introduced into an evacuated flask of 100 litre capacity at 610 K. Under the given conditions PCl_5 dissociates as $PCl_{5(g)} \rightleftharpoons PCl_{3(g)} + Cl_{2(g)}$ and the system is allowed to equilibrate. At equilibrium, the total pressure of mixture was found to be 6.0 atm. The K_p for the reaction is___$\times 10^{-2}$.

$\left[\text{Given: } R = 0.082 L\ atm K^{-1} mol^{-1}\right]$

50.



The graph above shows a best-fit line for a kinetic experiment on the reaction $A \rightarrow$ products. The best-fit straight line is

$$\frac{1}{[A]} = kt + \frac{1}{[A]_0},$$

Below are 7 statements about the reaction and the graph. How many of these statements are incorrect?

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Statements:

1) The reaction is second order with respect to A.

2) The integrated rate law is $\frac{1}{[A]} = kt + \frac{1}{[A]_0}$.

3) The unit of the rate constant k is s^{-1}.

4) The initial rate of disappearance of A (at t=0) equals $0.080\, mol\, L^{-1}s^{-1}$.

5) The half-life $t_{1/2}$ for this reaction (with the given $[A]_0$) equals $25.0\, s$.

6) The time required for $[A]$ to fall to $0.50\, M$ is $75.0\ s$.

7) If the initial concentration $[A]_0$ were doubled, the half-life $t_{1/2}$ would also double.

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. If z_1, z_2, z_3 are complex numbers represented by three distinct points lying on the circle $|z| = 2$, then the minimum value of $|z_1 + z_2|^2 + |z_2 + z_3|^2 + |z_3 + z_1|^2$ is equal to

A) 6 B) 12 C) 15 D) 24

52. Let $f : R \to R$ be a twice differentiable function such that

$(\sin x \cos y)(f(x+y) - f(x-y)) = (\cos x \sin y)(f(x+y) + f(x-y))$,

for all $x, y \in R$. If $f'(0) = 1$, then the value of $6f''\left(\frac{5\pi}{3}\right)$ is equal to

A) 3 B) $3\sqrt{3}$ C) $-3\sqrt{3}$ D) – 3

53. If S and S_1 are foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{16} = 1$ and P is any point on it, then the difference of maximum value and minimum value of $(SP)(S_1P)$ is equal to

A) 16 B) 9 C) 15 D) 25

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54. Let $L_1 : \vec{r} = \left(\hat{i}+\hat{j}+\hat{k}\right)+\lambda\left(\hat{i}+\hat{j}\right), \lambda \in R$ and $L_2 : \vec{r} = \hat{i}+(\mu+1)\hat{j}+(\mu+1)\hat{k}, \mu \in R$ be two lines intersecting at point 'A'.Through point $B(3,3,1)$ a line is drawn making an angle of 60^0 with L_2 and intersecting it at point C. Then the value of area of ΔABC is equal to

A) $2\sqrt{3}$ B) $2\sqrt{2}$ C) $4\sqrt{3}$ D) $4\sqrt{2}$

55. Let $P_n = \alpha^n + \beta^n, n \in N$. If $P_{15} = 1364, P_{14} = 843, P_{13} = 521$ and $P_1 = 1$, then the quadratic equation having roots $\frac{1}{\alpha} and \frac{1}{\beta}$ is

A) $x^2 - x + 1 = 0$ B) $x^2 + x - 1 = 0$

C) $x^2 - x - 1 = 0$ D) $x^2 + x + 1 = 0$

56. Given distinct straight lines OA and OB. From a point in OA a perpendicular is drawn to OB; from the foot of this perpendicular a line is drawn perpendicular to OA. From the foot of this second perpendicular a line is drawn perpendicular to OB; and so on indefinitely. The lengths of the first and second perpendiculars are 8 and 6 respectively. Then the sum of the lengths of the perpendiculars approaches a limit as the number of perpendiculars grows beyond all bounds. This limit is equal to

A) 34 B) 64 C) 17 D) 32

57. Let $b \in R$ and B be a 3×3 matrix such that $\det(B) = 5, B + I = \begin{bmatrix} 3 & b & 1 \\ 2 & 2 & 1 \\ 1 & 0 & 3 \end{bmatrix}$

If $\det\left(\left((b+1)(adjB)\right)\left((b+2)B\right)\right) = 2^p 3^q 5^r$ and $p, q, r \in \{-10, -9, ...0, 1, ..., 20\}$, then the value of $p+q+r$ is equal to

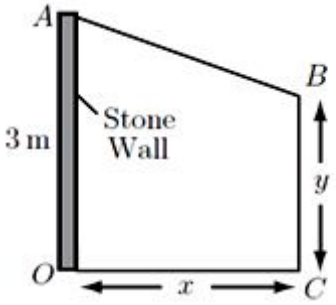
A) 14 B) 4 C) 7 D) 3

58. If two distinct chords of a parabola $y^2 = 4ax (a > 0)$ passing through the point (a, 2a) are bisected by the line $x + y = 1$, then the length of latus rectum of the parabola can not be equal to

A)1 B) 2 C) 3 D) 4

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59. A materials store is to be constructed next to a 3 metre high stone wall (shown as OA in the cross section in the diagram). The roof (AB) and front (BC) are to be constructed from alluminium metal sheeting. Only 6 metre length sheets are available. Each of them is to be cut into two parts such that one part is used for the roof and the other is used for the front. Let OC=x,BC=y.



If the cross -sectional area of material store is maximum then the value of greatest integer less than or equal to $2x^2+y^2$ is equal to

A) 34 B) 48 C) 24 D) 40

60. Let $A(\vec{a})$ and $B(\vec{b})$ be points on two skew lines $\vec{r}=\vec{a}+\lambda\vec{p}$ and $\vec{r}=\vec{b}+\mu\vec{q}$ where $\lambda,\mu\in R$ and the shortest distance between the skew lines is 1, where $\vec{p}$ and $\vec{q}$ are unit vectors forming adjacent sides of parallelogram whose area is $\frac{1}{2}$ square units. If an angle between AB and the line of shortest distance is 60°, then the value of $(AB)^2$ is equal to

A) $\frac{9}{4}$ B) 4 C) 1 D) $\frac{1}{4}$

61. Let $f_n(x)=\sum_{j=1}^{n}\cot^{-1}\left(1-(x+j)+(x+j)^2\right)$ for all $x\geq 0$. Then the value of $\sum_{j=1}^{10}\left(j^2+1\right)\sin^2\left(f_j(0)\right)$ is equal to

A) 3025 B) 385 C) 770 D) 6050

62. Let [t] denotes the greatest integer function. If $\int_0^1\left[1+x^2+x^4\right]dx=a$, then the value of $36a-25a^2+8a^3-a^4$ is equal to

A) 19 B) – 19 C) 18 D) – 21

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63. A circle concentric with the ellipse $\frac{x^2}{25}+\frac{y^2}{b^2}==1$ cuts the ellipse at four distinct points A, B, C, D. If e_1 is the eccentricity of the ellipse when ABCD is a rectangle having maximum area 40 square units and e_2 is the eccentricity of the ellipse when ABCD is a square of area $\frac{450}{17}$ square units, then the value of e_1+e_2 is equal to

A) 1 B) $\frac{8}{5}$ C) $\frac{6}{5}$ D) $\frac{7}{5}$

64. The value of $\lim\limits_{x\to 0}\frac{\sec x-(\sec x)^{\sec x}}{1-\sec x+\ln(\sec x)}$ is equal to

A) 0 B) 1 C) 2 D) – 1

65. Let A be the set of all functions $f:Z\to Z$ and R be a relation on A such that $R=\{(f,g): f(2)=g(3) and\ f(3)=g(2)\}$. Then R is

A) Symmetric and transitive but not reflective

B) Symmetric but neither reflexive nor transitive

C) Reflexive but neither symmetric nor transitive

D) Transitive but neither reflexive nor symmetric

66. Three real numbers $\lambda_1,\lambda_2,\lambda_3$ are in G.P. and $\lambda_1,\lambda_2,\lambda_3-1$ are in A.P. If λ_1 is such that the equations $x-y+z=2, 2x+2y+\lambda_1 z=7,\ 3x+\lambda_1 y+4y+z=12$ has infinite solutions, then which of the following is true?

A) $\lambda_1+2,\lambda_2+1,\lambda_3$ form an A.P.

B) $\lambda_1+2,\lambda_2+1,\lambda_3$ form a G.P.

C) Maximum and minimum values of the function $f(x)=\lambda_1\sin x+\lambda_2\cos x, x\in R$ are $\sqrt{3}$ and $-\sqrt{3}$ respectively.

D) Range of the function $g(x)=\lambda_1\tan^{-1}x+\lambda_3\cot^{-1}x, x\in R$ is $\left(\frac{\pi}{2},\frac{7\pi}{2}\right)$.

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67. Let a vector $\overline{V_1} = \overline{OP}$, where O is origin and P(2, 0, 2). Vector $\overline{V_1}$ is rotated about O by an angle of $\frac{\pi}{3}$ in a plane which is perpendicular to XOZ plane to get vector $\overline{V_2}$ where $\overline{V_2}.\hat{j} > 0$. Again $\overline{V_1}$ is rotated about origin by right angle in XOZ plane to get vector $\overline{V_3}$. Then the possible value of $\overline{V_2}.\hat{j} + \overline{V_3}.\hat{k} - 2$ is equal to

A) $\sqrt{6}$ B) $-\sqrt{6}$ C) 3 D) – 3

68. The solution of the differential equation $\frac{dy}{dx} = \frac{\sin y + x}{\sin 2y - x\cos y}$ is

A) $\sin^2 y = x \sin y + \frac{x^2}{2} + C$ B) $\sin^2 y = x \sin y - \frac{x^2}{2} + C$

C $\sin^2 y = x + \sin y + \frac{x^2}{2} + C$ D) $\sin^2 y = x - \sin y + \frac{x^2}{2} + C$

69. The sum of distinct solutions of the equation $\left(4\tan x + \tan^2 x + 1\right) = 2\sqrt{2}\sin\left(x + \frac{\pi}{4}\right)\left(1 + \tan^2 x\right)$ in the interval $[0, \pi]$ is equal to

A) $\frac{\pi}{2}$ B) $\frac{4\pi}{3}$ C) 2π D) $\frac{7\pi}{6}$

70. Let $\binom{n}{k}$ denote the number of ways of choosing k objects out of n distinct objects.

If $\sum_{k=1}^{20}\binom{20}{k}\binom{20}{k-1}(-1)^k + \sum_{k=0}^{20}\binom{20}{k}^2(-1)^k + \sum_{k=0}^{21}\binom{21}{k}^2(-1)^k = p\binom{19}{10}$, then $p^2 - p$ is equal to

A) 2 B) 0 C) 12 D) 6

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SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. Let $f(x)$ be a continuous function which takes positive values for all $x \geq 0$ and satisfy $\int_0^x f(t)\,dt = x\sqrt{f(x)}$ where $f(1) = \frac{1}{2}$.

If $f\left(\sqrt{2}+1\right) = \frac{a}{b}$ where a and b are positive coprime numbers, then the value of $a+b$ is equal to

72. Let two elements (a, b), (c, d) be selected randomly from the set $S = \{(m,n) : m,n \in \{1,2,.....,10\}, m \neq n\}$. If the probability that $a+b=c+d$ is p, then the value of $(45)^2 p$ is equal to:

73. The number of 6-digit numbers formed by using all the digits 1, 3, 4, 5, 6, 8 and divisible by 11, is equal to

74. Let there be two circles passing through (-1,1) and touching the lines $x+y=2$ *and* $x-y=2$ whose radii are r_1 and r_2. If $r_1 + r_2 = a\sqrt{2}$, then a is equal to

75. If the area of region $\left\{(x,y) \in R^2 : y \geq \sqrt{|x+3|}, 5y \leq x+9 \leq 15\right\}$ is equal to k then the value of 2k is equal to

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